

# CONSOLE BASED PROJECT – HOSPITAL MANAGEMENT

(OOPS CONCEPTS)

## CODE:

```
import java.util.*;

class Patient {

    private int patientId;

    private String patientName;

    private int age;

    public Patient(int patientId, String patientName, int age) {

        this.patientId = patientId;

        this.patientName = patientName;

        this.age = age;

    }

    public int getPatientId() {

        return patientId;

    }

    public String getPatientName() {

        return patientName;

    }

    public int getAge() {

        return age;

    }

    public double calculateBill() {

        return 0;

    }

}
```

```

class InPatient extends Patient {
    private int daysAdmitted;

    public InPatient(int patientId, String patientName, int age,
        int daysAdmitted) {
        super(patientId, patientName, age);
        this.daysAdmitted = daysAdmitted;
    }

    @Override
    public double calculateBill() {
        double roomCharge = daysAdmitted * 2000;
        double consultationFee = 500;
        double labCharge = 1000;
        return roomCharge + consultationFee + labCharge;
    }
}

class OutPatient extends Patient {
    public OutPatient(int patientId, String patientName, int age) {
        super(patientId, patientName, age);
    }

    @Override
    public double calculateBill() {
        double consultationFee = 500;
        double labCharge = 300;
        return consultationFee + labCharge;
    }
}

class Doctor {
    private int doctorId;
    private String doctorName;

```

```

private String specialization;

public Doctor(int doctorId, String doctorName,
              String specialization) {
    this.doctorId = doctorId;
    this.doctorName = doctorName;
    this.specialization = specialization;
}

public int getDoctorId() {
    return doctorId;
}

public String getDoctorName() {
    return doctorName;
}

public String getSpecialization() {
    return specialization;
}
}

class Appointment {
    private Patient patient;
    private Doctor doctor;

    public Appointment(Patient patient, Doctor doctor) {
        this.patient = patient;
        this.doctor = doctor;
    }

    public void displayAppointment() {
        System.out.println("\n----- APPOINTMENT DETAILS -----");
        System.out.println("Patient ID    : "
                           + patient.getPatientId());
        System.out.println("Patient Name : "

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        + patient.getPatientName());
System.out.println("Age      : "
        + patient.getAge());
System.out.println("Doctor ID   : "
        + doctor.getDoctorId());
System.out.println("Doctor Name  : "
        + doctor.getDoctorName());
System.out.println("Specialization : "
        + doctor.getSpecialization());
    }
}

public class Hospital {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Doctor[] doctors = {
            new Doctor(101, "Ramesh", "Cardiologist"),
            new Doctor(102, "Suresh", "Neurologist"),
            new Doctor(103, "Priya", "Dermatologist"),
            new Doctor(104, "Anitha", "Orthopedic")
        };
        while (true) {
            System.out.println("\n===== HOSPITAL MANAGEMENT =====");
            System.out.println("1. In Patient");
            System.out.println("2. Out Patient");
            System.out.println("3. Exit");
            System.out.print("Enter Choice : ");
            int choice = sc.nextInt();
            if (choice == 3) {
                System.out.println("Thank You");
            }
        }
    }
}

```

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        break;
    }
    System.out.print("Enter Patient ID : ");
    int id = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Patient Name : ");
    String name = sc.nextLine();
    System.out.print("Enter Age : ");
    int age = sc.nextInt();
    Patient patient;
    if (choice == 1) {
        System.out.print("Enter Days Admitted : ");
        int days = sc.nextInt();
        patient = new InPatient(id, name, age, days);
    } else {
        patient = new OutPatient(id, name, age);
    }
    System.out.println("\nChoose Specialization");
    for (int i = 0; i < doctors.length; i++) {
        System.out.println((i + 1) + ". "
            + doctors[i].getSpecialization()
            + " - Dr. "
            + doctors[i].getDoctorName());
    }
    System.out.print("Enter Choice : ");
    int doctorChoice = sc.nextInt();
    if (doctorChoice < 1 || doctorChoice > doctors.length) {
        System.out.println("Invalid Doctor Choice!");
        continue;
    }

```

```
}  
  
Doctor doctor = doctors[doctorChoice - 1];  
  
Appointment appointment =  
    new Appointment(patient,  
        doctor);  
appointment.displayAppointment();  
  
double bill =  
    patient.calculateBill();  
System.out.println(  
    "\nTotal Bill Amount : Rs."  
        + bill);  
}  
sc.close();  
}  
}
```

## OUTPUT:

```
===== HOSPITAL MANAGEMENT =====
1. In Patient
2. Out Patient
3. Exit
Enter Choice : 1
Enter Patient ID : 101
Enter Patient Name : Priya
Enter Age : 28
Enter Days Admitted : 2

Choose Specialization
1. Cardiologist - Dr. Ramesh
2. Neurologist - Dr. Suresh
3. Dermatologist - Dr. Priya
4. Orthopedic - Dr. Anitha
Enter Choice : 4

----- APPOINTMENT DETAILS -----
Patient ID      : 101
Patient Name    : Priya
Age            : 28
Doctor ID       : 104
Doctor Name     : Anitha
Specialization  : Orthopedic

Total Bill Amount : Rs.5500.0
```

```
===== HOSPITAL MANAGEMENT =====
1. In Patient
2. Out Patient
3. Exit
Enter Choice : 2
Enter Patient ID : 102
Enter Patient Name : Kathir
Enter Age : 34

Choose Specialization
1. Cardiologist - Dr. Ramesh
2. Neurologist - Dr. Suresh
3. Dermatologist - Dr. Priya
4. Orthopedic - Dr. Anitha
Enter Choice : 2

----- APPOINTMENT DETAILS -----
Patient ID      : 102
Patient Name    : Kathir
Age            : 34
Doctor ID       : 102
Doctor Name     : Suresh
Specialization  : Neurologist

Total Bill Amount : Rs.800.0

===== HOSPITAL MANAGEMENT =====
1. In Patient
2. Out Patient
3. Exit
Enter Choice : 3
Thank You
```